Alexander Ky, BS

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EDUCATION

M.D. Candidate with **Public Health Certificate** 08/2023 - 05/2027
Michigan State University – *College of Human Medicine (MSU-CHM)* East Lansing, MI
Electives: Musculoskeletal (MSK), Microbiology, Lifestyle Medicine, Anesthesiology

B.S. in Biochemistry, Minor in Applied Statistics 08/2018 - 04/2022
University of Michigan Ann Arbor, MI
Literature, Science, and the Arts - Honors

PUBLICATIONS

1. Singu VK, Nguyen NM, **Ky A**, Schaal VL, Athota P, Gowen A, VanOrmer M, Loh O, Blount AJ, Hanson C, Yelamanchili SV, Anderson-Berry A, Pendyala G. Dietary Supplementation with Omega-3 Fatty Acids Rescues Neurodevelopmental Deficits in a Combined Exposure Paradigm of Maternal Oxycodone Exposure and Chronic Stress. *J. Neuroimmune Pharmacol. Revision submitted* **2026**.

2. Kwon JJ, Li J, Liu JY, Patsalis C, Wu P, Kawase YH, **Ky A**, Pasternak C, Mason DD, Bozadjeva-Kramer N, Loebel C, Lumeng CN, O'Rourke RW. A Hydrogel Culture System Regulates Human Adipocyte Function. *Int. J. Mol. Sci.* **2025**, 26(22), DOI: 10.3390/ijms262210865.

3. Loonen S, Shah A, Michmerhuizen B, Decker B, **Ky A**, Park J, Dishay D, Krohmer J, Jones J, Lindermann E. 43 hazardous chemical accidents: A data-driven study of incidents in the United States during 2021-2024. *Ann Emerg Med.* **2025**, 86(3S1); S17-S18. DOI: 10.1016/j.annemergmed.2025.06.054

4. **Ky A**, McCoy AJ, Flesher CG, Friend NE, Li J, Akinleye K, Patsalis C, Lumeng CN, Putnam AJ, O'Rourke RW. Matrix density regulates adipocyte phenotype. *Adipocyte* **2023,** 12(1), DOI: 10.1080/21623945.2023.2268261.

5. Kang Q, Zhu X, Ren D, **Ky A**, MacDougald OA, O'Rourke RW, Rui L. Adipose METTL14-elicited N6-methyladenosine promotes obesity, insulin resistance, and NAFLD through suppressing β adrenergic signaling and lipolysis. *Advanced Science* **2023**, 10(28), DOI: 10.1002/advs.202301645.

6. Strieder-Barboza C, Flesher CG, Geletka LM, Eichler T, Akinleye O, **Ky A**, Ehlers AP, Lumeng CN, O'Rourke RW. Lumican modulates adipocyte function in obesity-associated type 2 diabetes. *Adipocyte* **2022,** 11(1), DOI: 10.1080/21623945.2022.2154112.

7. Carruthers NJ, Strieder-Barboza C, Caruso JA, Flesher CG, Baker NA, Kerk SA, **Ky A**, Ehlers AP, Varban OA, Lyssiotis CA, Lumeng CN, Stemmer PM, O'Rourke RW. The human type 2 diabetes-specific visceral adipose tissue proteome and transcriptome in obesity. *Sci Rep* **2021**, 11(1):17394, DOI: 10.1038/s41598-021-96995-0.

PRESENTATIONS AND SCHOLARLY WORKS

1. **Ky A,** Nallamothu N, Shukri O, Hediyal TA, Stone EM, Schaal VL, Foroughi-Nezhad A, Davis JS, Yelamanchili SV, Pendyala G. From Nerve Relief to Ovarian Grief: Gabapentin-Induced Reproductive Toxicity. Accepted for poster presentation at ASA Anesthesiology Annual Conference, San Diego, CA, Oct. 25, 2026.

2. Vichaikul S, Tsai M, Reynoso JG, **Ky A**, Ablir R, Choi S, Trottier M. A community-based study: Understanding mental health help-seeking behavior and barriers among Southeast Asian Americans in Michigan. Poster presentation at APAMSA National Conference, San Francisco, CA, Feb. 20, 2026.

3. Loonen S, Shah A, Michmerhuizen B, Decker B, **Ky A**, Park J, Dishay D, Krohmer J, Jones J, Lindermann E. 43 hazardous chemical accidents: A data-driven study of incidents in the United States during 2021-2024. Corewell Health 2025. Oral presentation at ACEP25 Research Forum, Salt Lake City, UT, Nov. 7, 2025.

4. Kwon, J, **Ky A**, Li J, Chai B, Akinleye K, O'Rourke RW. Matrix density regulates adipocyte phenotypes. Poster presentation at University of Michigan Department of Surgery Moses Gunn Research Conference, Ann Arbor, MI, May 30, 2025.

5. **Ky A**, Bennett N, Judge B, Courtley M, Jones J. Atropa belladonna (deadly nightshade) intoxication: A case report. Corewell Health 2024. Poster presentation at Pediatric Great Lakes Conference, East Lansing, MI, Mar. 13, 2025.

6. Jerrick T, Chassee T, Rauch L, **Ky A**. The use of high flow nasal cannula during interhospital transfers in pediatric patients. Corewell Health Helen DeVos Children's Hospital 2024. Poster presentation at National Association of EMS Physicians, San Diego, CA, Jan. 9, 2025.

7. **Ky A,** Li J, Akinleye K, Kwon J, Lumeng CN, O'Rourke RW. A machine-learning based algorithm for quantifying in vitro adipocyte differentiation efficiency. Poster presentation at University of Michigan Department of Surgery VA Research Week, Ann Arbor, MI, May 13, 2024.

8. **Ky A.** RSV Patient Education. Poster Presentation at Scholarly Project Presentation Day, Michigan State University College of Human Medicine, Grand Rapids, MI, Feb. 27, 2024.

9. **Ky A,** Geletka L, Eichler T, Akinleye O, Flesher CG, Delproposto D, Lumeng CN, O'Rourke RW. Adipose tissue stromal cells regulate systemic metabolism in a depot specific manner. Oral presentation at University of Michigan Department of Surgery Moses Gunn Research Conference, Ann Arbor, MI, May 26, 2022.

10. Strieder-Barboza C, Flesher CG, Geletka LM, Delproposto JB, Eichler T, Akinleye O, **Ky A**, Ehlers AP, O'Rourke RW, Lumeng CN. Single-nuclei transcriptome of human adipose tissue reveals metabolically distinct depot-specific adipose progenitor subpopulations. *bioRxiv* **2022**, DOI: 10.1101/2022.06.29.496888. *Preprint.*

11. **Ky A**, Bonebright A, Greineder C. Targeted drug delivery to treat ischemia-reperfusion injury. University of Michigan Biointerfaces Institute. Poster presentation at UROP Annual Spring Research Symposium, Ann Arbor, MI, April 24, 2019.

RESEARCH

Medical Student Anesthesia Research Fellowship (MSARF) 06/2026 - 08/2026
University of Nebraska – Department of Anesthesia Omaha, NE
Principal Investigator: Gurudutt Pendyala, PhD
Ongoing projects: Effects of gabapentin on uterine and ovarian architecture and development in pregnant rats; Validation of AI models for pre-operative anesthesia decision making

Senior Research Project 08/2025 - Present
Hurley Medical Center Flint, MI
Principal Investigator: Rasheed Hassan, MD
Ongoing Project: The impact of preoperative peripheral nerve blocks on postoperative pain and opioid use in patients undergoing lower extremity amputations

O'Rourke Lab 01/2021 - 05/2025
University of Michigan Department of General Surgery – *Research Lab Technician* Ann Arbor, MI
Principal Investigator: Robert O'Rourke, MD
Ongoing Project: Using machine learning to quantify adipocyte differentiation

Pediatric Emergency Medicine Research Group (PEMRG) 02/2024 - 01/2026
Corewell Health Emergency Medicine – *Research Assistant* Grand Rapids, MI
Principal Investigators: Jeffery Jones, MD and Tonia Jerrick, MD
Project: The use of high flow nasal cannula during interhospital transfers in pediatric patients

Trinity Health Neurology 10/2023 - 01/2025
Trinity Health – *Research Assistant* Grand Rapids, MI
Principal Investigator: Aileen Antonio, MD
Project: Obstructive sleep apnea education for Asian Americans in West Michigan

Greineder Lab 10/2018 - 04/2019
University of Michigan, Ann Arbor, Biointerfaces Institute – *Research Assistant* Ann Arbor, MI
Principal Investigator: Colin Greineder, MD
Project: Targeted drug delivery to treat ischemia-reperfusion injury

EXPERIENCES

Health Record Grader 05/2026 - Present
- Graded H&P and SOAP notes written by medical students during their simulation lab and worked with them on how to improve in preparation for their clinical rotations

Anatomy Lab Instructor 10/2024 - 03/2025
- Helped prepare cadavers by labeling key structures for anatomy lab
- Held open lab hours for students to come and ask questions

Nucleus Tutoring - *Tutor* 02/2024 - 04/2026

- Provided support with subjects such as Algebra, Chemistry, and Physics for high school and college students

Student Admissions Ambassador 08/2023 - 04/2024

- Interviewed medical school applicants
- Conducted tours of the medical school campus and participated in panels for incoming medical students

Quantum Physics for Kids - *Teacher* 10/2022 - 01/2025

- Used a gaming platform to teach chemistry fundamentals such as atomic structures and circuit making

West Ottawa Science Olympiad - *Event Coach* 01/2022 - 05/2025

- Taught middle school students anatomy and physiology, epidemiology, and chemistry in preparation for competitions

IHA West Arbor Pediatrics, Ann Arbor - *Medical Assistant* 01/2021 - 07/2023

- Performed procedures such as vaccinations, suture removals, and administration of medications while maintaining medical records accurately through EPIC and Michigan Care Improvement Registry (MICR)

Michigan Men's Rowing Team, Ann Arbor - *Coxswain* 09/2019 - 12/2022

- Trained with freshmen rowers as a coxswain to develop leadership experience

University of Michigan Housing, Ann Arbor - *Residential Advisor* 08/2019 - 04/2021

- Guided over 50 freshmen through class scheduling and internship interviews
- Developed multiple programs with strong turnouts that were based on diversity, equity, and inclusion values

AWARDS

Alpha Omega Alpha (AOA) Medical Honor Society 2026

- Awarded to 20% from each class in recognition of outstanding academic record, leadership, scholarship, and professionalism

Gold Humanism Honor Society 2026

- Awarded to 15% from each class for demonstrating patient-centered care by modeling integrity, excellence, compassion, altruism, respect and empathy

Medical Student Anesthesia Research Fellowship (MSARF) 2026

- Competitive, nationally awarded fellowship supporting medical student research in anesthesiology

Florence Gravelle Oberg Endowed Memorial Scholarship 2026

Spartan Volunteer Service Award 2025

- Over 100 volunteer hours of direct community engagement and services

West Michigan Medical Student Scholarship 2024

The Richard and Helen Devos Scholarship 2024

Health Policy Case Competition 2022
Awards: 1st Place and Anti-Racism and Economic Mobility Award
- Explored ways to improve healthcare access and reducing health disparities in Michigan by addressing social determinants of health as well as developing policy to implement it

Phi Sigma Kappa National President's Special Achievement Award for Campus Involvement 2019

Phi Sigma Kappa Delta Deuteron Scholarship 2019

Order of Omega Case Study Competition 2019
Award: 2nd Place
- Investigated how to address scenarios involving greek-life on campus by proposing to use restorative practices rather than punitive measures to deal with situations

LEADERSHIP POSITIONS

MSU-CHM Continuous Quality Improvement (CQI) Committee 01/2026 - Present
- Offered advice on improvement of student resources during monthly meetings with administration

Lifestyle Medicine Interest Group (LMIG) – *Vice President* 04/2024 - 02/2025
- Created an elective to teach medical students the core pillars of lifestyle medicine
- Organized events to help students and the community become active and engaged in a healthy lifestyle

Asian Pacific American Medical Student Association (APAMSA) – *Social Chair* 08/2023 - 02/2025
- Attended conferences and presentations regarding Asian Pacific Americans in medicine to learn about disparities and current research
- Fostered a welcoming environment for incoming medical students

Grand Rapids Surgery Interest Group (GRSIG) - *Officer* 08/2023 - 02/2025
- Led suturing elective to medical students to help them prepare for rotations
- Went out into the community to teach college students suture techniques and to answer questions about medical school

Emergency Medicine Interest Group (EMIG) - *Officer* 08/2023 - 02/2025
- Hosted events to promote the field of emergency medicine
- Invited EM doctors to come speak to students to share what it is like in the emergency room

Side Door, Ann Arbor - *Co-Founder* 01/2023 - 08/2023
- Researched recipes and expanded cooking repertoire through collaborations and classes
- Shared cultural understandings through home cooked foods with a cultural twist

University of Michigan Housing - *Resstaff Coordinator* 08/2021 - 05/2022
- Oversaw 40 residential advisors to create an inclusive and well-run dormitory
- Advised hall council to create events and promote community within the dorm
- Addressed concerns and feedback with upper housing to help improve Resstaff

Phi Sigma Kappa Fraternity - *Delta Deuteron Chapter Secretary* 01/2019 - 01/2022

- Maintained accurate chapter records and attendance
- Participated in philanthropy events to support Special Olympics

Asian American Association - *Freshman Representative* 09/2018 - 05/2019

- Planned events that promoted cultural diversity and community building
- Recruited freshmen to engage with ongoing campus events

COMMUNITY SERVICE AND ADVOCACY

Wellness Services - *Volunteer* 04/2026 - Present

- Delivered harm reduction care through a needle exchange program, including wound assessment and basic treatment for injection-related infections
- Provided patient-centered education on overdose prevention and safer substance use

Michigan State Medical Society 02/2026 - Present

- Wrote and sponsored a resolution to make preventative and routine care transportation accessible, reliable, and patient-centered in Michigan

Battambang Provincial Referral Hospital – Volunteer 05/2025 - 06/2025

- Assisted with patient triage, discharge education, and patient communication

Grand Rapids Rowing Club - *Novice Coach* 07/2024 - 05/2025

- Fostered a supportive and collaborative environment to enhance athletes' confidence, discipline, and sportsmanship
- Coached novice and intermediate rowers on proper rowing techniques, stroke mechanics, and teamwork strategies to improve performance

Hope Clinic - *Clinical Assistant* 06/2022 - 03/2023

- Volunteered once a week to help establish care for underserved communities
- Worked with social workers and other specialists to provide well-rounded care

PROFESSIONAL AFFILIATIONS

American Society of Anesthesiologists (ASA)
American Medical Association (AMA)
Asian Pacific American Medical Student Association (APAMSA)
Michigan State Medical Society (MSMS)

SKILLS AND CERTIFICATION

Advanced Cardiovascular Life Support (ACLS) American Heart Association

Basic Life Support (BLS) American Heart Association

Collaborative Institute Training Initiative (CITI)

- Social-Behavioral Human Subjects Researcher
- MSU Graduate School Responsible Conduct of Research (**RCR**)
- GCP for Social and Behavioral Investigators and Staff
- Biomedical Research Investigators and Key Personnel

University of Michigan

- Responsible Conduct of Research and Scholarship (**RCRS**) Training

Software

EPIC Systems

Healthcare Software; Completed Research Fundamentals Training

Cerner – Oracle Health

Electronic Health Record

REDCap

Electronic Data Capture (EDC) software for clinical & translational research

RStudio

Statistics Software

MiChart

Electronic Health Record System

Microsoft Excel, Word & PowerPoint

Microsoft Office Programs

Statistical Package for the Social Sciences (SPSS)

Data Analysis Software

Languages

English – Native Fluency

Khmer – Native Fluency